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Guidance for Antimicrobial Prophylaxis for Dental Procedures after Joint Replacement Surgery

Introduction: Dental prophylaxis after total joint arthroplasty is not recommended by the American Academy of Orthopaedic Surgeons (AAOS) or the American Dental Association (ADA). However, data are limited and some patient populations may benefit from this practice. This guidance is intended to delineate the suspected population of benefit.

General Prevention: Preferably, patients should have good dental care prior to surgery. For patients without dental care, dental referral should be provided as soon as possible, including upon referral to orthopedic surgery for consideration of surgical management to optimize outcomes. Routine dental care should be avoided in the three months after surgery, but best practice would be to undergo dental evaluation in the few months prior to surgery.

Additional rationale and data for the above can be found on page 3.

Table 1: When to consider antimicrobial prophylaxis for patients with prosthetic joints prior to dental procedures. Potential harms of prescription should also be considered*. See Table 2 for preferred agents/doses.

Any of the following for patients undergoing routine dental cleaning: - Previous prosthetic joint infection with any organism - Severe Immunocompromise, including any of: o HIV with CD4 < 200   o Neutropenia with ANC < 1000   o History of Solid organ transplant   o Allogeneic stem cell transplant < 1 year ago or with GVHD   o Hematologic malignancy (active)   o Immunosuppressive medications (excluding hydroxychloroquine, methotrexate and steroids of < 20 mg daily)   o Steroids pred ≥20 mg daily (or equivalent)   o Asplenia   o Active chemotherapy - The patient is unlikely to be a candidate for further procedures in the opinion of the surgeon due to global surgical risk and/or challenging local characteristics of the joint. (e.g., Tumor prosthesis) - Meet any other established criteria for dental prophylaxis (e.g., Prosthetic heart valve) *The above criteria apply if met when receiving dental care, not only at time of joint surgery.*
Patients with poor dentition on examination or no dental care in the last 12 months who are undergoing routine cleaning, if they are < 12 months from joint replacement surgery
Patients who do not meet these criteria, particularly those who are < 12 months out from joint replacement, who are requesting dental prophylaxis should be engaged in a risk benefit discussion* of this practice and if they feel the potential benefits outweigh the potential risks, prescription is reasonable.

*Benefits of dental prophylaxis include potentially decreasing the chance of PJI after dental work.

Risks include adverse effects from antibiotics including rash/allergy, diarrhea, nausea/vomiting, stomach upset and *C. difficile* infection. It is not clear that prophylaxis decrease the risk of PJI or that dental procedures are associated with PJI and even in cases of oral organism PJI thought to be from dental procedure, prophylaxis is not 100% effective.

Table 2: Agents for dental prophylaxis

	Drug	Dose
1st line	**Amoxicillin**	2000 mg by mouth one hour prior to procedure ONCE
For patients with low-risk allergy to amoxicillin (see Michigan Medicine Beta Lactam Allergy Guide)	**Cephalexin**	2000 mg by mouth one hour prior to procedure ONCE
If high risk allergy to amoxicillin or allergy to both amoxicillin and cephalexin (See Michigan Medicine Beta Lactam Allergy Guide)	**Doxycycline**	100 mg by mouth one hour prior to procedure ONCE

Note: Clindamycin has high rates of adverse effects and antibiotic resistance and is no longer recommended as prophylaxis

Background:
The American Academy of Orthopaedic Surgeons' (AAOS)[1] and American Association of Hip and Knee Surgeons (AAHKS) and American Dental Association's[2] (ADA) most recent guidelines, endorsed by the Infectious Diseases Society of American (IDSA) and the Musculoskeletal Infection Society (MSIS) do not recommend the prescription of dental prophylaxis for all patients after joint replacement surgery. This change was made because the available data do not suggest an association between dental procedures and prosthetic joint infection (PJI)[3-6]. Additionally, most PJI are caused by organisms that are not typically found in the oropharynx. Streptococcal species account for 7-12% of PJI[7]. Furthermore, while many studies do not distinguish oral streptococci from all streptococci, the available data[8] suggest only 20-25% of cases of streptococcal PJI are likely to be from the oropharynx (i.e., 2-4% of all PJI). In addition, transient bacteremia with oral streptococci is commonly seen after routine everyday activities such as tooth brushing, flossing and chewing[9] and is not exclusive to dental procedures. However, high quality evidence as to the utility of antimicrobial prophylaxis in patients with prosthetic joints is lacking and due to the high morbidity associated with PJI, there may be populations that benefit from this practice, particularly in the first 12 months after joint replacement when risk of PJI is the highest[10, 11]. This guidance is intended to help delineate which populations may most benefit from dental prophylaxis after joint replacement's performed by Michigan Medicine surgeons.

Dental Care Prior to surgery:
Dental health is an important part of overall health and when possible, patients should be routinely assessed for access to dental care and referred when able. Good dental hygiene is also protective against complications after arthroplasty, including but not limited to infection[11]. Thus, all patients under consideration for joint replacement surgery would ideally be under the care of a dentist and have routine dental care both prior to and after joint replacement surgery. If a patient may be a candidate for joint replacement surgery and they are not under dental care, they should be counseled to seek this out and when possible, referred to a dentist. This step should occur as early in the potential joint replacement process as possible, including by referring providers prior to or concurrently with referral to orthopedic surgery for consideration of surgical management. In patients who have not had routine dental care, delaying joint replacement to allow for dental examination and procedures if necessary should be considered. However, recognizing that access to dental care is not universal and that there are cases in which waiting for dental examination may lead to prolonged morbidity, delaying surgery for this will not be appropriate in all circumstances. Obtaining and maintaining good dental hygiene and engaging in regular dental care is more protective than intermittent prophylaxis and is preferred. Due to potential risks, it is advised to avoid routine dental care in the 3 months immediately after surgery, however dental examination in the few months prior to surgery to correct or prevent any dental needs is considered best practice.

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References

	Citation
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Revision History

Date	Revision

The recommendations in this guide are meant to serve as treatment guidelines for use at Michigan Medicine facilities. If you are an individual experiencing a medical emergency, call 911 immediately. These guidelines should not replace a provider's professional medical advice based on clinical judgment, or be used in lieu of an Infectious Diseases consultation when necessary. As a result of ongoing research, practice guidelines may from time to time change. The authors of these guidelines have made all attempts to ensure the accuracy based on current information, however, due to ongoing research, users of these guidelines are strongly encouraged to confirm the information contained within them through an independent source